Dongki Jung

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EDUCATION

University of Maryland, College Park
- Ph.D. in Computer Science — Aug 2024 – Present
  - Adviser: Prof. Dinesh Manocha

Korea Advanced Institute of Technology (KAIST)
- M.S. in Electrical Engineering — Feb 2019 – Feb 2021
  - Adviser: Prof. Changick Kim

Korea University
- B.S. in Electrical Engineering — Mar 2013 – Feb 2019
  - Auxiliary Police (mandatory military service) — May 2014 – Feb 2016

EMPLOYMENT

Research Internship at Meta Reality Labs
- Develop Asset Generation techniques for high-fidelity mesh reconstruction. — Jun 2026 – Present

Research Scientist at NAVER LABS — Apr 2021 – Aug 2024
- Developed Digital Twin techniques and VR tour pipelines for real estate platforms.
- Developed Neural Rendering algorithms for Augmented Reality platform.

Research Internship at NAVER LABS
- Developed Robot Perception algorithms for Visual Navigation. — Sep 2020 – Mar 2021

RESEARCH INTERESTS

3D Foundation & Generative Models, and Traditional SfM

PUBLICATIONS

PREPRINTS

[1] Heechan Yoon, **Dongki Jung**, Phuc Nguyen, Ming Lin, Dinesh Manocha, "PanoSeg3R: Feed-Forward 3D Semantic Segmentation for Panoramic Images with an Automatic Data Curation Pipeline", *Submitted.*

[2] Adil Qureshi, **Dongki Jung**, Jaehoon Choi, Dinesh Manocha, "PanoPlane: Plane-Aware Panoramic Completion for Sparse-View Indoor 3D Gaussian Splatting", *Submitted.*

[3] Phuc Nguyen, Xiyi Chen, **Dongki Jung**, Anshul Rai, Guan-Ming Su, Dinesh Manocha, Ming Lin, "Scalable 3D Reconstruction and Understanding", *Submitted.*

[4] Jaehoon Choi, **Dongki Jung**, Yonghan Lee, Sungmin Eum, Dinesh Manocha, and Heesung Kwon, "UAVTwin: Neural Digital Twins for UAVs using Gaussian Splatting", *Submitted.*

INTERNATIONAL CONFERENCES

[1] **Dongki Jung**, Jaehoon Choi, Adil Qureshi, Somi Jeong, Dinesh Manocha, and Suyong Yeon, "Wid3R: Wide Field-of-View 3D Reconstruction via Camera Model Conditioning", ***ECCV***, 2026

[2] **Dongki Jung**, Jaehoon Choi, Yonghan Lee, Dinesh Manocha, "MoRe: Monocular Geometry Refinement via Graph Optimization for Cross-View Consistency", ***WACV***, 2026.

[3] Jaehoon Choi, **Dongki Jung**, Christopher Maxey, Sungmin Eum, Yonghan Lee,Dinesh Manocha, and Heesung Kwon, "UAV4D: Dynamic Neural Rendering of Human-Centric UAV Imagery using Gaussian Splatting", ***AAAI***, 2026

[4] **Dongki Jung**, Jaehoon Choi, Yonghan Lee, and Dinesh Manocha, "RPG360: Robust 360 Depth Estimation with Perspective Foundation Models and Graph Optimization", ***NeurIPS***, 2025.

[5] **Dongki Jung***, Jaehoon Choi*, Yonghan Lee, Dinesh Manocha, "IM360: Textured Mesh Reconstruction for Large-scale Indoor Mapping with 360° Cameras," ***ICCV***, 2025. (* equal contribution)

[6] **Dongki Jung**, Jaehoon Choi, Yonghan Lee, Somi Jeong, Taejae Lee, Dinesh Manocha, Suyong Yeon, "EDM: Equirectangular Projection-Oriented Dense Kernelized Feature Matching," ***CVPR***, 2025.

[7] Obin Kwon, **Dongki Jung**, Youngji Kim, Soohyun Ryu, Suyong Yeon, Songhwai Oh, Donghwan Lee, "WayIL: Image-based Indoor Localization with Wayfinding Maps," ***ICRA***, 2024.

[8] Jaehoon Choi, **Dongki Jung**, Taejae Lee, Sangwook Kim, Youngdong Jung, Dinesh Manocha, Donghwan Lee, "TMO: Textured Mesh Acquisition of Objects with a Mobile Device by using Differentiable Rendering," ***CVPR***, 2023.

[9] **Dongki Jung***, Jaehoon Choi*, Yonghan Lee, Deokhwa Kim, Dinesh Manocha, Donghwan Lee, "SelfTune: Metrically Scaled Monocular Depth Estimation through Self-Supervised Learning," ***ICRA***, 2022. (* equal contribution)

[10] **Dongki Jung***, Jaehoon Choi*, Yonghan Lee, Deokhwa Kim, Changick Kim, Dinesh Manocha, Donghwan Lee, "DnD: Dense Depth Estimation in Crowded Indoor Dynamic Scenes," ***ICCV***, 2021. (* equal contribution)

[11] Taekyung Kim, Jaehoon Choi, Seokeon Choi, **Dongki Jung**, Changick Kim, "A Few Depth Points are All You Need for Multi-view Stereo: A Novel Semi-supervised Learning Method for Multi-view Stereo," ***ICCV***, 2021.

[12] Jaehoon Choi, **Dongki Jung**, Yonghan Lee, Deokhwa Kim, Dinesh Manocha, and Donghwan Lee, "SelfDeco: Self-Supervised Monocular Depth Completion in Challenging Indoor Environments," ***ICRA***, 2021.

[13] **Dongki Jung***, Jaehoon Choi*, Donghwan Lee, Changick Kim, "SAFENet: Self-Supervised Monocular Depth Estimation with Semantic-Aware Feature Extraction," ***NeurIPSW***, 2020. (* equal contribution)

[14] **Dongki Jung**, Seunghan Yang, Jaehoon Choi, and Changick Kim, "Arbitrary Style Transfer Using Graph Instance Normalization," ***ICIP***, 2020.

[15] Yonghan Lee, Jaehoon Choi, **Dongki Jung**, Jaeseong Yun, Soohyun Ryu, Dinesh Manocha, Suyong Yeon, "Mode-GS: Monocular Depth Guided Anchored 3D Gaussian Splatting for Robust Ground-View Scene Rendering," *arXiv* 2024.

[16] Seunghan Yang, Youngeun Kim, **Dongki Jung**, Changick Kim, "Partial Domain Adaptation Using Graph Convolutional Networks," *arXiv* 2020.

CHALLENGES

INTERNATIONAL CHALLENGES

[1] **3rd place** in the Track 3: City-Scale Multi-Camera Vehicle Tracking at **AI City Challenge** held in *IEEE Conference on Computer Vision and Pattern Recognition* Workshop 2020

PATENTS

[1] Obin Kwon, Suyong Yeon, Soohyun Yoo, Youngji Kim, **Dongki Jung**, Donghwan Lee, "Method, Computer Device, and Computer Program for Localization with Wayfinding Map," Korean Patent No. 10-2023-0186934

[2] **Dongki Jung**, Taejae Lee, Donghwan Lee, Suyong Yeon, "Method for Estimating Depth based on Equirectangular Image and Computer Device using the same," Korean Patent No. 10-2023-0159197

[3] Gihwan Lee, **Dongki Jung**, Taejae Lee Donghwan Lee, "Method for Generating 3D Mesh based on Equirectangular Image and Computer Device using the same," Korean Patent No. 10-2023-0160309

[4] Taejae Lee, **Dongki Jung**, Gihwan Lee, Jaesung Park, Youngho Jeong, Donghwan Lee, Suyong Yeon, Sangwook Kim, Moonyong Choi, "Method, Apparatus, System and Computer Program for Scanning Space Image and Providing Virtual Reality Service," Korean Patent No.10-2023-0169485

[5] **Dongki Jung**, Taejae Lee, Donghwan Lee, Jaeseong Yun, "Method for updating texture map of 3D mesh and computing device using the same," Korean Patent No. 10-2023-0153382

[6] Jaesung Park, Taejae Lee, Gihwan Lee, **Dongki Jung**, Donghwan Lee, SangWook Kim, Suyong Yeon, "Method for Rendering Grid and Apparatus thereof," Korean Patent No. 10-2023-0130352

[7] Taejae Lee, **Dongki Jung**, Gihwan Lee, Donghwan Lee, "Method for generating indoor floor plan based on image and computing device using the same," Korean Patent No. 10-2023-0177654

[8] Taejae Lee, **Dongki Jung**, Suyong Yeon, Donghwan Lee, "Vision based Localization Method and Apparatus thereof," Korean Patent No. 10-2023-0169736

[9] **Dongki Jung**, Donghwan Lee, Yonghan Lee, Deokhwa Kim, "Method and System for Training Monocular Depth Estimation Models," Korean Patent No. 10-2023-0064188

TEACHING

- University of Maryland College Park, Teaching Assistant
  - CMSC351 – Algorithms
  - CMSC216 – Introduction to Computer Systems
  - CMSC421 – Introduction to Artificial Intelligence

AWARDS & SCHOLARSHIPS

- Academic Achievement Award, Korea University
  - Semester High Honors in the first Semester of 2016
  - Semester High Honors in the second Semester of 2016
  - Semester High Honors in the first Semester of 2017
  - Semester High Honors in the second Semester of 2017
  - Great Honor in Winter 2018 Graduation
- KU Alumni Scholarships
  - the second Semester of 2016
- YooJung Scholarship Foundation
  - the first and second Semesters of 2017
  - the first and second Semesters of 2018

LANGUAGES

- Korean (Native), English (Fluent)

SKILLS

Python, C++, ROS, Docker, LaTeX, MATLAB, PyTorch, TensorFlow,

REFERENCES

- **Dinesh Manocha**
  Professor of Computer Science and Professor of Electrical and Computer Engineering
  University of Maryland, College Park
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- **Changick Kim**
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